SPEAR GUY SURVIVAL

Introduction

Spear Guy Survival is a 2D game where the user will face a boss in battles similar to the souls-like genre, wherein users must have good timing and decision-making skills. One of the biggest challenges when making a game boss is making sure that the boss' actions are unpredictable enough for players not to predict them easily. In our game, we made use of a Monte Carlo Tree Search inspired mechanism for the boss to adapt to battle scenarios..

Key Ideas:

- Built in GameMaker Studio 2 with fast-paced melee mechanics
- Player uses movement, attacks, and blocking to survive
- Boss has multiple attack options with different timing and range
- Uses Monte Carlo Tree Search decision-making
- Attack choices are selected using weighted probabilities
- Weights update based on success (hit) vs failure (miss)
- Results in a boss that adapts over time instead of following fixed patterns

Boss AI: MCTS-Inspired Decision Making

Goal

Make the boss less predictable and more adaptive than a fixed attack pattern.

How does it work?

- Action Set: Slash, Thrust, Upward Thrust, Counter
- Selection: The boss chooses one action using weighted random selection.
- Update Rule: Successful attacks increase their weight, while unsuccessful attacks decrease it.

Why does it help?

The boss gradually favors actions that work better against the player, which makes the fight feel more dynamic and less scripted.

DEMO: MCTS-Inspired Decision Making

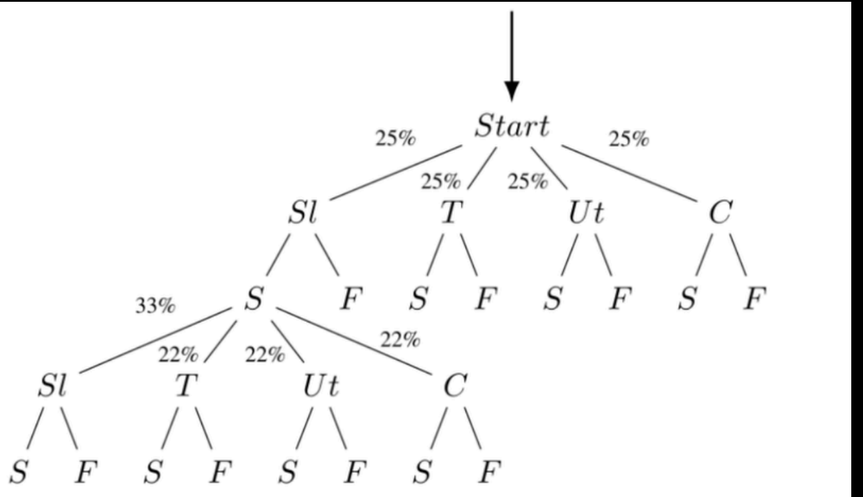

Figure 1: Enemy Decision Tree